1 Multi-Grasp PPO Training

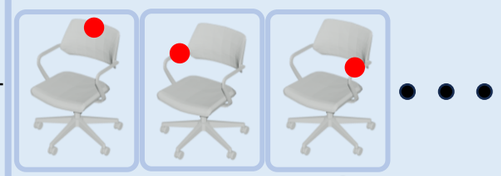
(Train a unified policy across multiple grasps)

Task observation
(robot, object, goal)



+

Initial grasps g_i



g1 (backrest) g2 (right armrest) g3 (left armrest)

High-level PPO policy (grasp-conditioned)

Actor
$\pi(a_t^{high} | o_t^{high}, g_i)$

Critic
$V^{\pi}(o_t^c(g_i), p_{goal}^{obj})$

Low-level controller
(pretrained locomotion + arm PD control)



Simulator (IsaacLab)
Train across grasp candidates g_i and goal pose p_{goal}^{obj}

2 Critic Distillation

(Learn a grasp utility network)

PPO critic (teacher)
$V^{\pi}(o^c(g_i), p_{goal}^{obj})$

Privileged critic observations
- Object state (e.g. velocity)
- Full robot state
- Grasp configuration g_i
- Environment info

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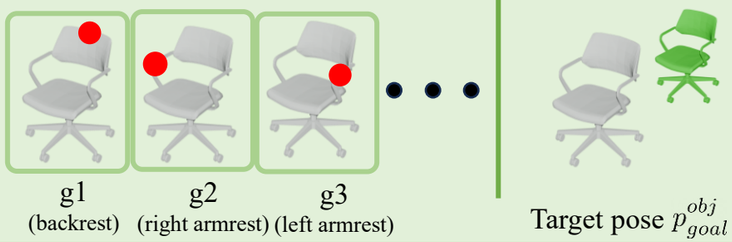
Grasp utility network (student)
$V_{grasp}(g_i, p_{goal}^{obj})$

Non-privileged observations
- Grasp configuration g_i
- Target object pose p_{goal}^{obj}

3 Task-Aware Grasp Selection and Execution

(Select the best grasp and execute the unified policy)

Candidate grasps g_i and target pose

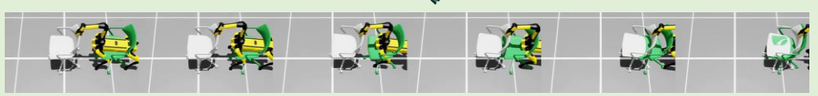


g1 (backrest) g2 (right armrest) g3 (left armrest) Target pose p_{goal}^{obj}

Grasp utility network
$V_{grasp}(g_i, p_{goal}^{obj})$

$$g^* = \arg\max_{g_i} V_{grasp}(g_i, p_{goal}^{obj})$$

Unified loco-manipulation policy
$\pi(a_t^{high} | o_t^{high}, g^*)$



Execute rearrangement (Simulator)